Sunny Pandey

Third Year Undergraduate
B.Tech Computer Science and Engineering (AI & DS)
MIT World Peace University, Pune

sunnysantoshpandey@gmail.com
sunnysantoshpandey
+91 93739 86194

Examination	Degree/Certificate	Institute	Year	CGPA/%
Graduation	B.Tech CSE (AI & DS)	MIT World Peace University, Pune	2024–Present	**9.5/10**
Intermediate/+2	Class XII	Taywade Junior College, Nagpur	2024	86.83%
Matriculation	Class X	G H Raisoni Vidyaniketan	2022	96.4%

Work Experience

Space Technology Research Group (STeRG) — *Sep'24 – Present*

Software Engineer, Embedded Systems Team : MIT-WPU, Pune

- Develop **flight and ground software** for the STeRG CubeSat: OBC flight software on **NASA cFS**, CFDP file transfer and the payload AI pipeline.
- Developed the **flight software for a CanSat** in Embedded C on **STM32**, covering real-time data acquisition and system monitoring.
- Interfaced **BNO055, BME280, INA219, L89 GNSS** and VEML sensors with STM32 over I2C and UART using HAL drivers.

MAAI — *Jul'25 – Present*

Founder : Pune

- Founded a startup building **MAAI**, an AI-powered wearable that monitors the vital signs of elderly patients and **alerts caregivers** in real time.

IRIS — *Present*

Tech Lead : MIT-WPU, Pune

- Lead the technical work of a team that builds **autonomous Pixhawk + Raspberry Pi UAVs**.
- Currently working on **Soteria**, the team's autonomous UAV platform.

Projects

THEIA: Onboard Image Screening Pipeline — *Jul'26*

Raspberry Pi CM5, 2× Coral Edge TPU, TensorFlow Lite, Python

- Built a real-time image screening pipeline on a CM5 with **two Coral Edge TPUs**: image quality and cloud assessment models (under 8 MB combined) share the first, the application model runs on the second.
- Classified each image as **passed, cropped or rejected** from its quality score and cloud coverage, so only useful images reach the application model.
- Ran capture and both inference stages as **separate processes** linked by bounded queues, monitored from a PyQt5 dashboard.

ARES: GPS-Denied Search and Rescue Drone — *Jul'26*

Smart India Hackathon 2026 : ROS 2, Pixhawk 6C, Raspberry Pi 5

- Designed an autonomous indoor drone that maps hazardous buildings with **LiDAR SLAM** and navigates using **frontier exploration** and **A* path planning**.
- Ran **YOLOv8n** on a Raspberry Pi 5 with Hailo-8L to detect survivors and tag their grid cell on the live map; fused optical flow and ToF in **ArduPilot EKF3** for position hold without GPS.
- Streamed the map, video and survivor markers to the ground station over a **private offline network**, with no cloud dependence.

OTA File Transfer — *May'26*

CCSDS CFDP Class 2, NASA CF, Python

- Implemented acknowledged file delivery between MCC, OBC and payload with **CFDP Class 2**, carrying PDUs inside ordinary CCSDS space packets in the telecommand and telemetry streams.
- Chained two transactions per transfer to **uplink AI models** to the payload computer and **downlink images**, with NAK-based retransmission, checksum verification and retry timers.
- Verified a Class 2 round trip between MCC and OBC on the bench link with **byte-identical files**.

Projects (contd.)

DAA-BOY: Handheld Game Console — *Apr'26*

PBL-II, Design and Analysis of Algorithms : STM32F446RE, C++

- Built a Game Boy-style console on an **STM32F446RE** with an ILI9341 TFT game screen, SSD1306 OLED HUD, joystick input and DFPlayer Mini audio.
- Implemented the WordQuest game on core algorithms: **AVL tree** search, **BFS** word ladder, **backtracking** with binary-search pruning and **LCS dynamic programming**.
- Added a merge-sorted leaderboard and a linear-probing hash table; kept the dictionary in Flash to fit within **128 KB of RAM**.

OBC Flight Software — *Mar'26*

NASA cFS, C, Raspberry Pi 5/CM5, Linux

- Built the on-board computer flight software on **NASA core Flight System** running on Linux on a Raspberry Pi CM5, with CFDP, stored-command, limit-checker and health-and-safety services.
- Wrote **custom cFS apps** that own each interboard link (EPS, ADCS, UHF, S-Band, payload) over **RS485 UART and TCP**, plus UDP command ingest and telemetry output.
- Implemented a **spacecraft mode manager** (BOOT, DETUMBLE, SAFE, NOMINAL, PAYLOAD_OPS, DOWNLINK) with autonomous battery-low-to-SAFE transitions.
- Unified every subsystem link onto one framed envelope with **CRC-8/DVB-S2**, replacing a per-link mix of XOR and SMBus checksums.

MAAI: Medical AI Alert Interface — *Aug'25*

Smart India Hackathon : STM32F446RE, ESP32, FreeRTOS

- Built a **wearable remote health monitor** for elderly care that tracks heart rate, SpO2, ECG, temperature and motion using MAX30102, AD8232, MLX90614 and MPU6050.
- Delivered a working **ESP32 + FreeRTOS** prototype with **fall detection** and RTOS task scheduling; now porting it to STM32F446RE with an LTE/GPS module for caregiver alerts.
- Added **camera-based monitoring** to detect falls and unusual behaviour alongside the sensor data.

BadhKavach: AI Flood Mitigation System

CNN, Image Segmentation

- Created a **segmentation-map generator** from terrain data to train CNN models for flood-risk zone detection and protection design.

Publications

- **U-Net Water Body Segmentation for CubeSat Deployment**

Technical Skills

- **Languages:** C, C++, Python
- **Embedded:** STM32, ESP32, Arduino, Raspberry Pi 5/CM5, FreeRTOS, HAL drivers, UART/RS485, I2C, SPI, ADC, DMA
- **Flight Software:** NASA cFS/cFE, CCSDS Space Packets, CFDP, OpenC3 COSMOS
- **AI & Robotics:** ROS 2, SLAM, Nav2, YOLOv8, TensorFlow Lite, Coral Edge TPU, Hailo-8L, OpenCV
- **Tools:** STM32CubeIDE, VS Code, Git, Linux
- **Spoken:** English (fluent), Hindi (native), Marathi (basic)